

CS4048 — DATA SCIENCE



Lecture 24 & 25

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FAST – NUCES, CFD Campus

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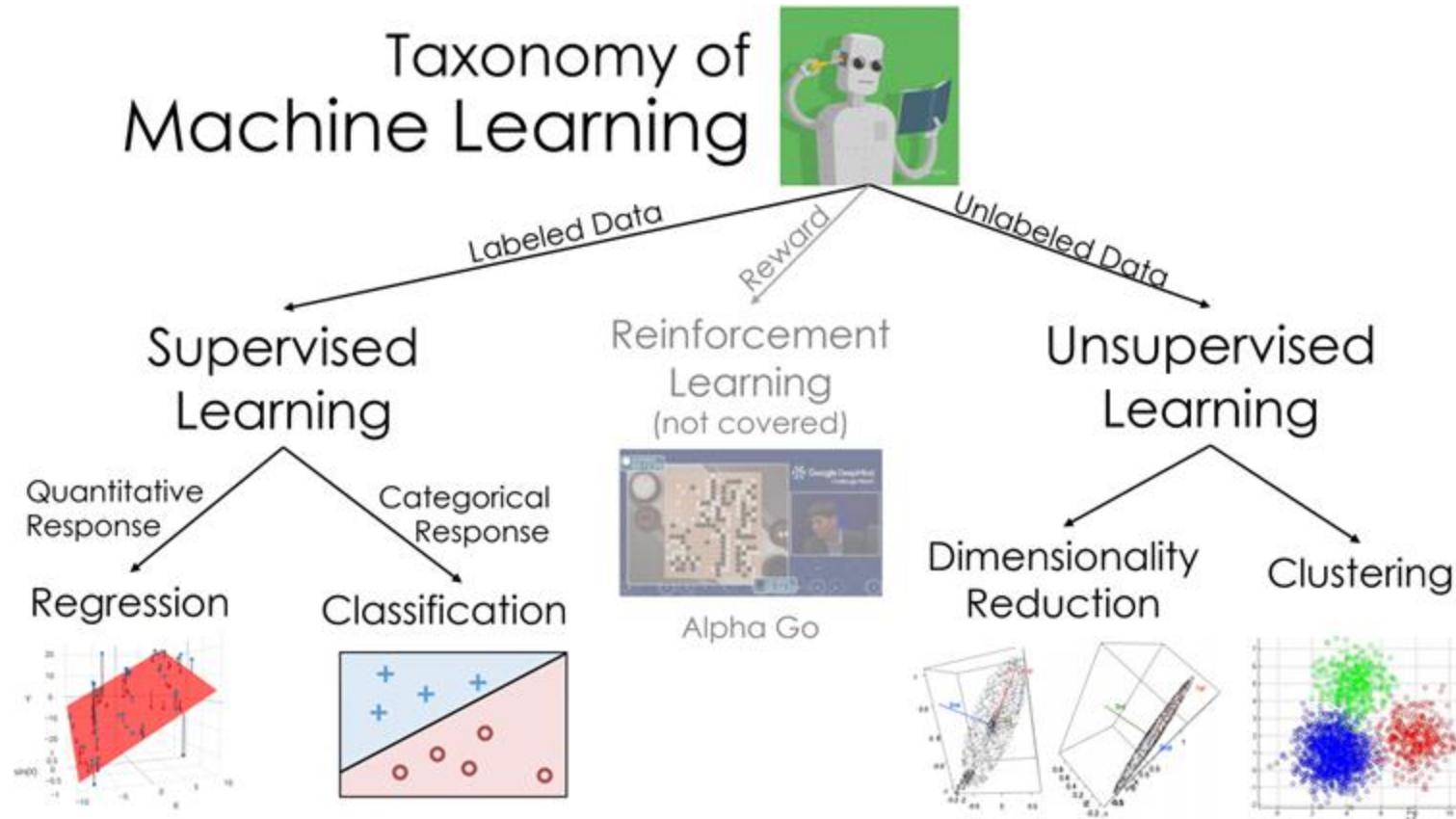
TODAY'S TOPICS

- Regression vs. Classification in ML
- Logistic Regression
- Cross entropy
- Evaluation metrics

BEYOND REGRESSION

Up until this point, we have been working exclusively with **linear regression**.

In the machine learning “landscape,” there are many other types of **learning problems**:

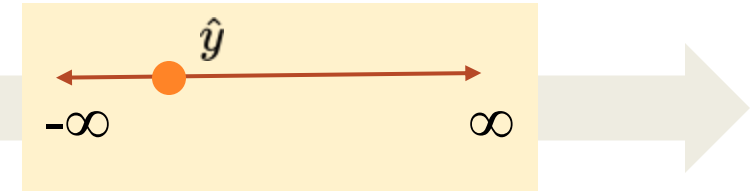


SO FAR: REGRESSION

In regression, we use quantitative features to predict a **numeric unbounded output**.

GAME_ID	TEAM_NAME	MATCHUP	REB	FTM	TOV	GOAL_DIFF	WON
21700001	Boston Celtics	BOS @ CLE	46	19	12	-0.049	0
21700002	Golden State Warriors	GSW vs. HOU	41	19	17	0.053	0
21700003	Charlotte Hornets	CHA @ DET	47	23	17	-0.030	0
21700004	Indiana Pacers	IND vs. BKN	47	25	14	0.041	1
21700005	Orlando Magic	ORL vs. MIA	50	22	15	0.042	1

$$x^T \theta$$



Input: numeric features

Model: linear combination

Output: numeric prediction

Examples:

- Predict **tip** from total bill
- Predict **mpg** from hp
- Predict **goal difference** from turnover %

NOW: CLASSIFICATION

In **classification**, we use quantitative features to predict a **qualitative (categorical) output**.

Examples:

- Predict **day of week** from total bill
- Predict **model of car** from hp
- Predict **which team won** from turnover %

We model **probabilities** of each class and transform them into a predicted **outcome**:

	TEAM_NAME	MATCHUP	REB	FTM	TOV	GOAL_DIFF	WON
GAME_ID							
21700001	Boston Celtics	BOS @ CLE	46	19	12	-0.049	0
21700002	Golden State Warriors	GSW vs. HOU	41	19	17	0.053	0
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21700004	Indiana Pacers	IND vs. BKN	47	25	14	0.041	1
21700005	Orlando Magic	ORL vs. MIA	50	22	15	0.042	1

$$p = \sigma(x^T \theta)$$

Win?
If $p \geq 0.5$: Predict a win
If $p < 0.5$: Predict a loss

$\hat{W}ON=0$
or
 $\hat{W}ON=1$

Input: numeric features

Model: linear
combination
transformed by **sigmoid**

Decision rule

Output:
**Predicted
class**



KINDS OF CLASSIFICATION

We want to predict some **categorical variable**, or **response**, y .

☒ **Binary** classification [\[we'll cover this\]](#)

- Two classes
- **Responses** (y) are either 0 or 1

win or lose

disease or
no disease

spam or
ham



Multiclass classification

- Many classes
- Image labeling and sentiment analysis (positive, negative, or neutral).



Structured prediction tasks

- Predicting a structured object instead of a discrete class
- Examples: Translation and ChatGPT.

THE MODELING PROCESS

Regression ($y \in \mathbb{R}$)

1. Choose a model

Linear Regression

$$\hat{y} = f_{\theta}(x) = x^T \theta$$

2. Choose a loss function

Squared Loss or
Absolute Loss

3. Fit the model

Regularization
Sklearn/Gradient descent

4. Evaluate model performance

R^2 , Residuals, etc.

Classification ($y \in \{0, 1\}$)

??

??

Regularization
Sklearn/Gradient descent

??
(Next lecture!)





LOGISTIC REGRESSION

LOGISTIC REGRESSION

- We use logistic **regression** to perform **classification**.
- Here, “regression” refers to fitting a function mapping inputs to outputs, not the task being performed.

THE GAMES DATASET

New modeling task, new dataset.

The games dataset describes the win/loss results of basketball teams.

GAME_ID	TEAM_NAME	MATCHUP	WON	GOAL_DIFF
21700001	Boston Celtics	BOS @ CLE	0	-0.049
21700002	Golden State Warriors	GSW vs. HOU	0	0.053
21700003	Charlotte Hornets	CHA @ DET	0	-0.030
21700004	Indiana Pacers	IND vs. BKN	1	0.041
21700005	Orlando Magic	ORL vs. MIA	1	0.042

Difference in field
goal success rate
between teams



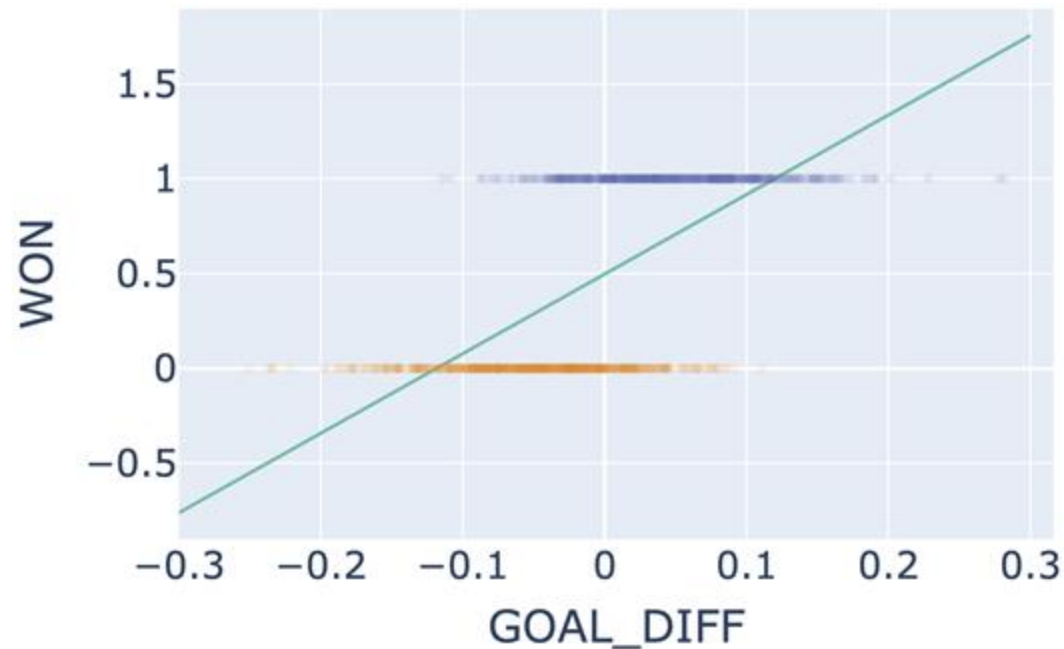
If a team won their game, then WON=1



WHY NOT OLS?

I suppose it is tempting, if the only tool you have is a hammer, to treat everything as if it were a nail.

– Abraham Maslow, *The Psychology of Science*

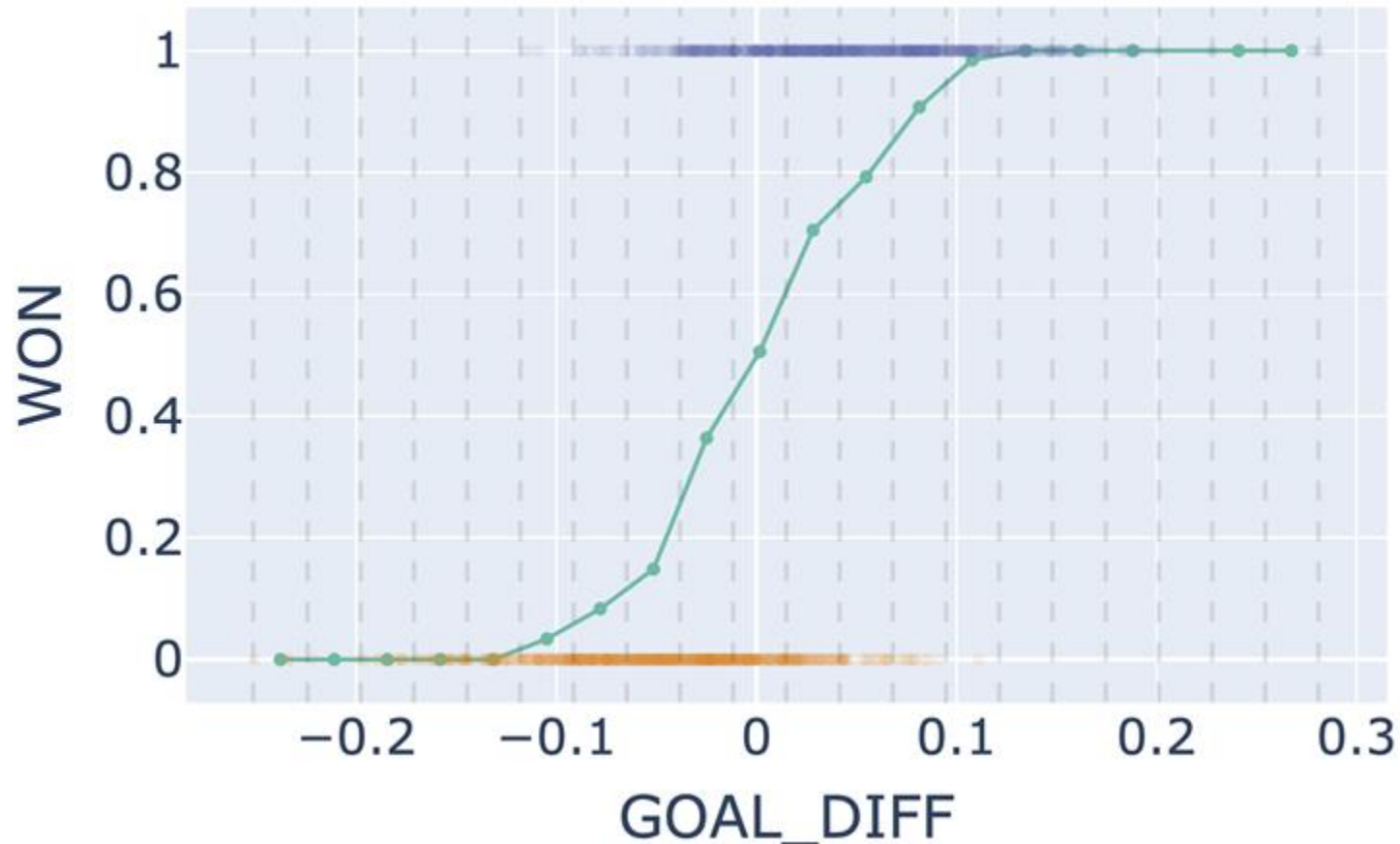


Problems:

- The output $\hat{y}$ can be outside the label range $\{0, 1\}$.
- Some outputs can't be interpreted: What does a class of "-0.6" mean?

TRANSFORMING REAL-VALUED FEATURES INTO [0,1] PROBABILITIES

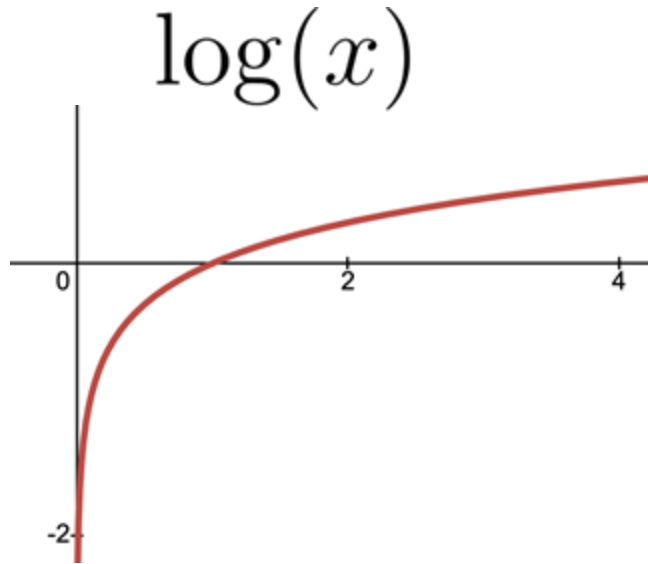
- We need to identify a function that maps $(-\infty, \infty)$ inputs to $(0,1)$ outputs.



LOG(X) MAPS $(0, \infty)$ INPUTS TO $(-\infty, \infty)$ OUTPUTS

$$\lim_{x \rightarrow \infty} \log(x) = \infty$$

As x increases, $\log(x)$ also increases without bound

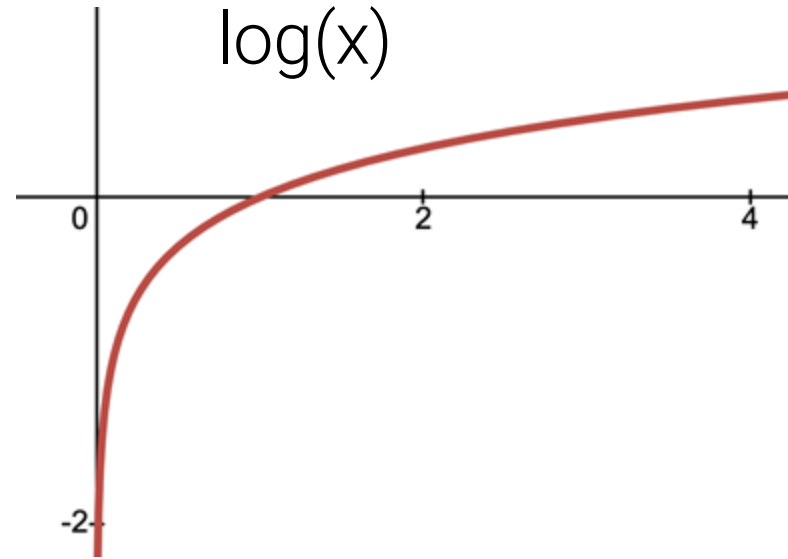


$$\lim_{x \rightarrow 0^+} \log(x) = -\infty$$

As x approaches 0 (from the positive side), $\log(x)$ decreases without bound

Also note that the logarithm is strictly increasing. Thus, $\log(x)$ maps $(0, \infty)$ to the range **$(-\infty, \infty)$**

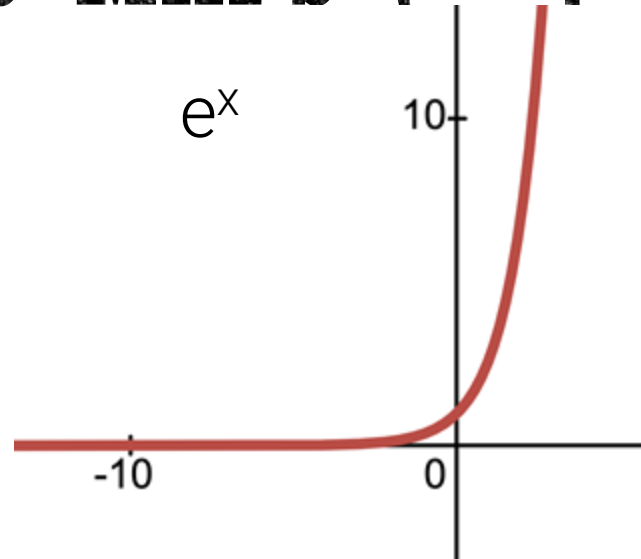
WHICH FUNCTIONS MAPS $(-\infty, \infty)$ INPUTS TO $(0,$



$$\lim_{x \rightarrow -\infty} \log(x) = \text{Undefined}$$

$$\lim_{x \rightarrow \infty} \log(x) = \infty$$

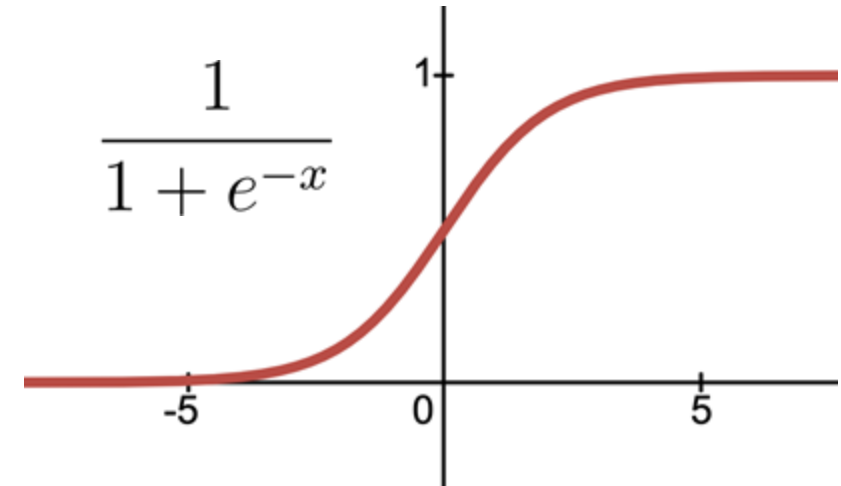
Not S-shaped, and
undefined for negative x .



$$\lim_{x \rightarrow -\infty} e^x = \lim_{x \rightarrow \infty} \frac{1}{e^x} = 0$$

$$\lim_{x \rightarrow \infty} e^x = \infty$$

Not S-shaped, maps
positive x to values greater
than 1 (thus, not in $[0, 1]$)



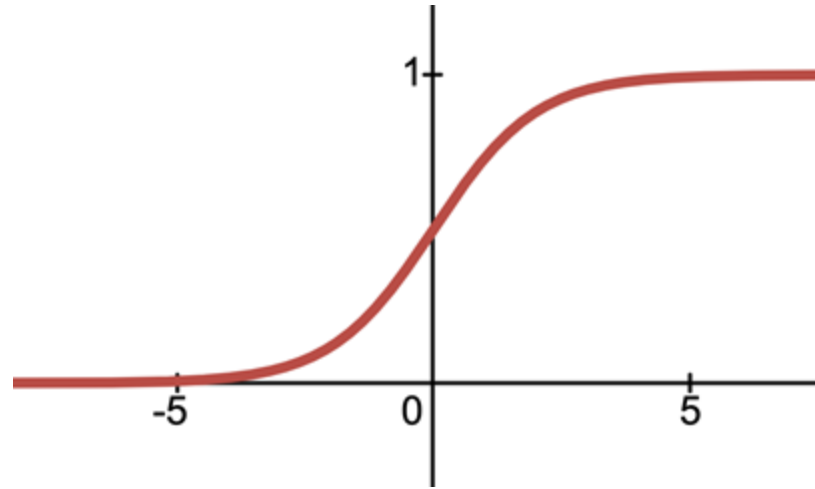
$$\lim_{x \rightarrow -\infty} \frac{1}{1 + e^{-x}} = 0$$

$$\lim_{x \rightarrow \infty} \frac{1}{1 + e^{-x}} = 1$$

**S-shaped, maps $-\infty$ to 0
and ∞ to 1.**

THE SIGMOID (σ) FUNCTION

Also known as the **logistic** function or the inverse logit function.

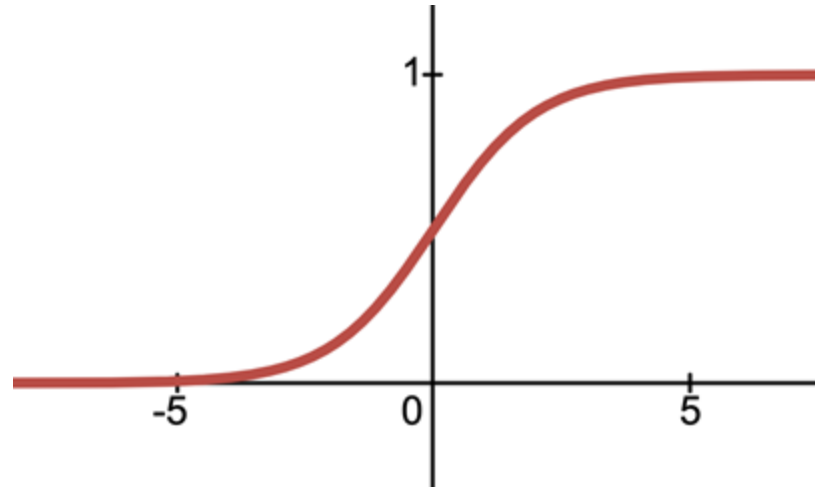


$$\sigma(x) = \frac{1}{\underbrace{1 + e^{-x}}_{}} = \frac{e^x}{1 + e^x}$$

We will default to this formulation of the sigmoid function.

TRANSFORMING THE SIGMOID (σ) FUNCTION

We can **shift** and **scale** the sigmoid function just like any other function:



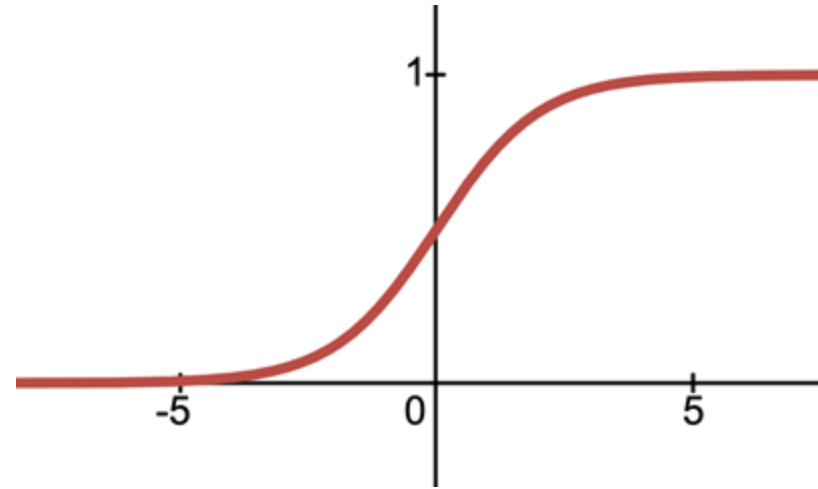
$$\sigma(Ax + B) = \frac{1}{1 + e^{-(Ax+B)}}$$

Bigger A $\rightarrow$ Horizontal shrink. Steeper!

Bigger B $\rightarrow$ Negative horizontal shift

TRANSFORMING THE SIGMOID (σ) FUNCTION

We can **shift** and **scale** the sigmoid function just like any other function:



Looks like SLR!

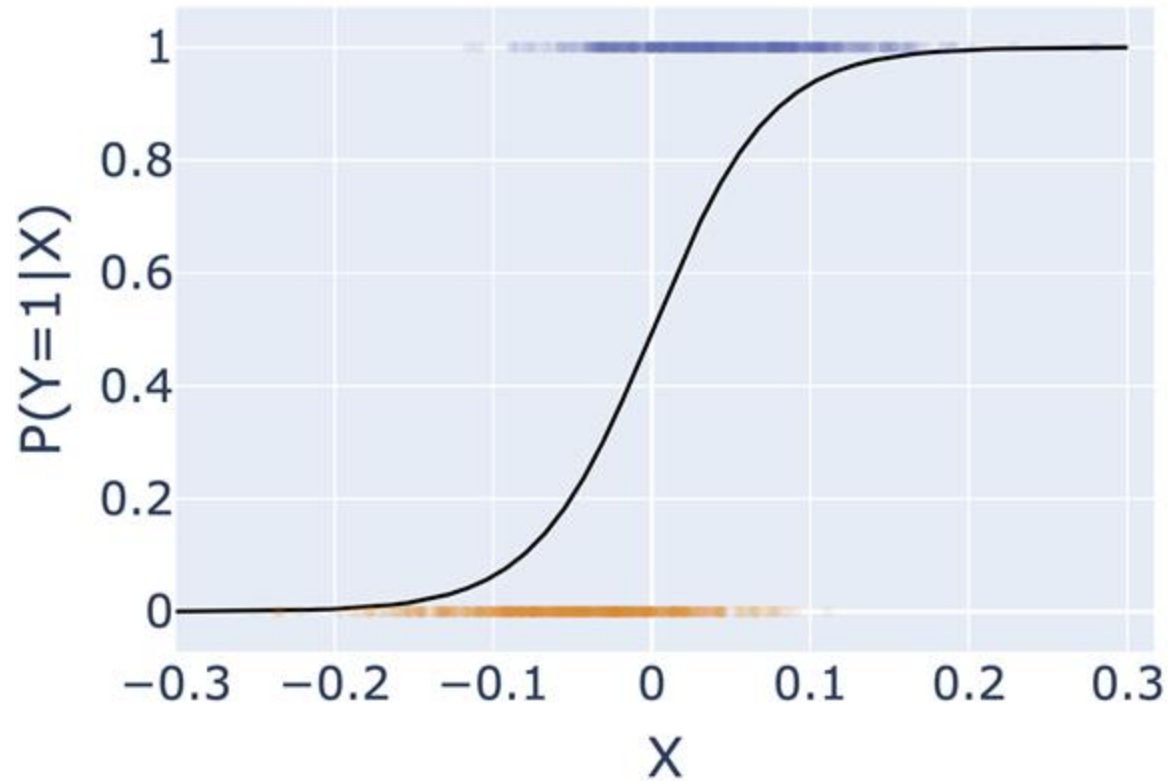
$$\sigma(\theta_0 + \theta_1 x) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x)}}$$

([Desmos](#))

Bigger $\theta_1 \rightarrow$ Horizontal shrink. Steeper!
Bigger $\theta_0 \rightarrow$ Negative horizontal shift.

"SIMPLE LOGISTIC REGRESSION"

We've discovered the (simple) **logistic regression** model! It just fits a sigmoid to the data.



$$P(Y = 1 \mid x) = \sigma(\theta_0 + \theta_1 x) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x)}}$$

How do we identify the optimal θ 's? More shortly!

"SIMPLE LOGISTIC REGRESSION"

Consider a **logistic regression model** with one feature and an intercept term ([Desmos](#)):

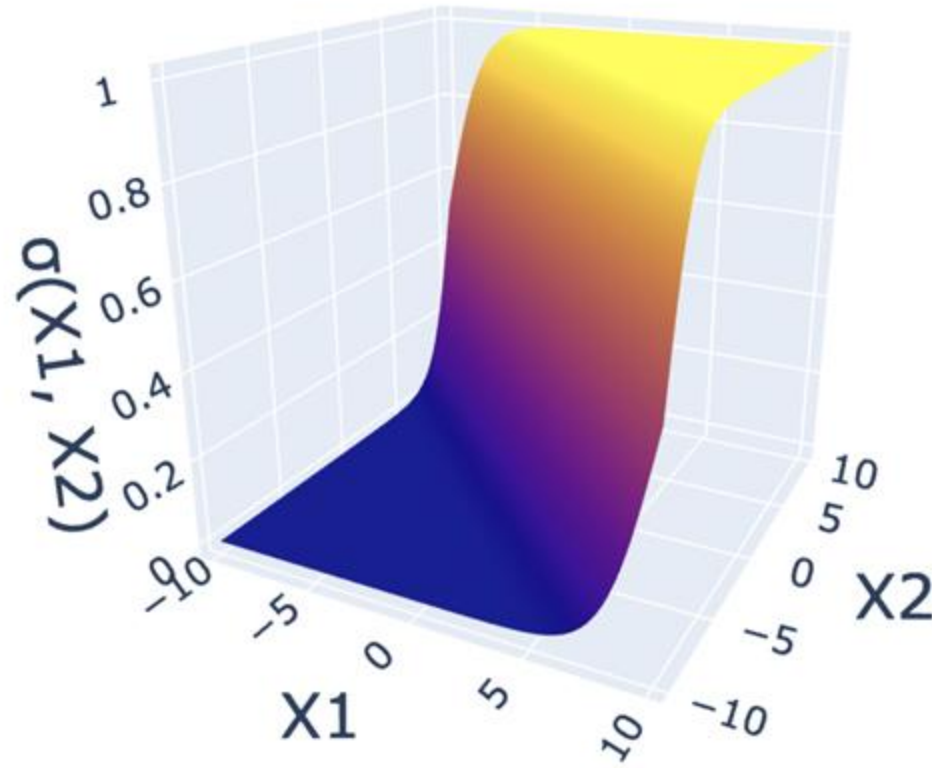
$$P(Y = 1 | x) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x)}}$$

θ_0 and θ_1 transform the default sigmoid function:

- θ_0 controls the **position** of the curve along the horizontal axis.
- The magnitude of θ_1 controls the **steepness** of the sigmoid.
- The sign of θ_1 controls the **orientation** of the curve.

LOGISTIC REGRESSION WITH TWO INPUTS

Just like OLS, we're not limited to one input:



$$P(Y = 1 \mid x) = \sigma(\theta_0 + \theta_1 x_1 + \theta_2 x_2) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x_1 + \theta_2 x_2)}}$$

LOGISTIC REGRESSION WITH P INPUTS

The **logistic regression model** for modeling $P(Y=1 | \mathbf{x})$:

$$\begin{aligned} P(Y = 1 | x) &= \frac{1}{1 + e^{-x^\top \theta}} = \sigma(x^\top \theta) \\ &= \frac{1}{1 + e^{-(\theta_0 + \theta_1 x_1 + \dots + \theta_p x_p)}} \end{aligned}$$

To predict a probability:

- Compute a **linear** combination of the features: $x^\top \theta$
- Apply the **sigmoid function**: $\sigma(x^\top \theta)$

COMPUTING ESTIMATED PROBABILITY FOR A NEW DATA POINT

I fit a logistic regression model **with no intercept** using my training data, and estimate these optimal parameters:

$$\hat{\theta}^T = [0.1 \quad -0.5]$$

$$P(Y = 1 \mid x) = \frac{1}{1 + e^{-x^\top \theta}}$$

What is the the estimated probability of $Y=1$ for a single observation where $x_0 = 15$ and $x_1 = 1$?

COMPUTING ESTIMATED PROBABILITY FOR A NEW DATA POINT

I fit a logistic regression model **with no intercept** using my training data, and estimate the optimal parameters:

$$\hat{\theta}^T = [0.1 \quad -0.5]$$

What is the the estimated probability of $Y=1$ if $x_0=15$ and $x_1 = 1$?

$$\begin{aligned}\hat{P}_{\hat{\theta}}(Y = 1|x) &= \sigma(x^T \hat{\theta}) \\ &= \sigma(0.1 \cdot 15 + (-0.5) \cdot 1) \\ &= \sigma(1) \\ &= \frac{1}{1 + e^{-1}} \\ &\approx 0.7311\end{aligned}$$

Because the response is more likely to be 1 than 0, we might predict $\hat{y} = 1$

(More on this decision in a bit!)

FORMALIZING THE LOGISTIC REGRESSION MODEL

The main takeaways of this section:

- Model the **probability** $P(Y = 1 | \mathbf{x})$ using a **sigmoid** transformation of a **linear model**.

$$\underbrace{\hat{P}_{\theta}(Y = 1|x)}_{\text{Estimated probability that the response is 1, given the features } x} = \underbrace{\frac{1}{1 + e^{-x^T \theta}}}_{\text{Sigmoid function } \sigma(x^T \theta) \text{ at the value } x^T \theta}$$

Estimated probability that the response is 1, given the features x

Sigmoid function $\sigma(x^T \theta)$ at the value $x^T \theta$

The logistic regression model is often written as follows:



$$\hat{P}_{\theta}(Y = 1|x) = \sigma(x^T \theta)$$

Looks like linear regression. Now wrapped with $\sigma()$!

Two Qs: (1) How do we **interpret** the θ 's? (2) How do we **estimate** the θ 's?



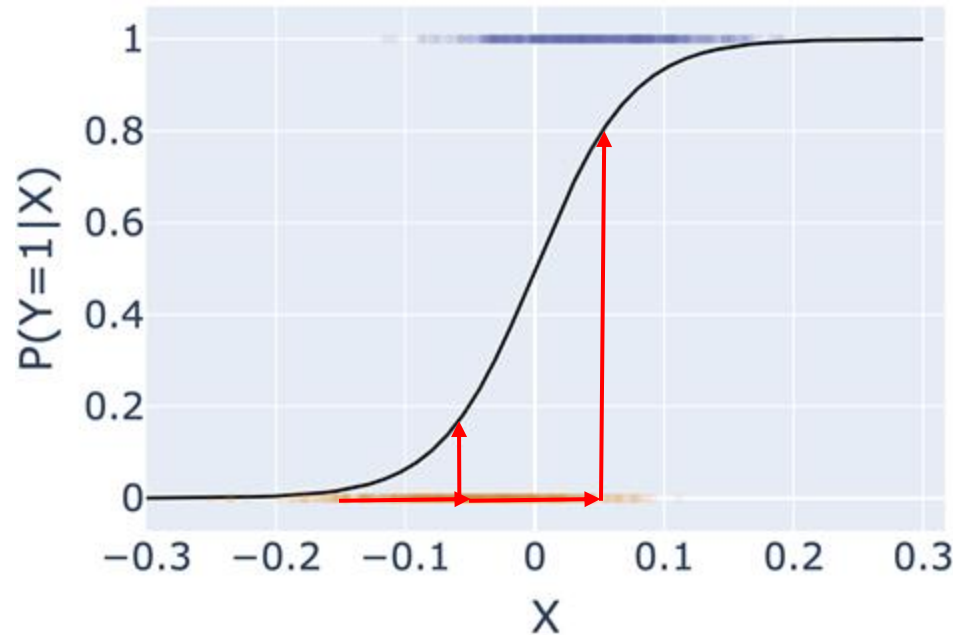
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LOG ODDS AND LINEARITY

INTERPRETING LOGISTIC REGRESSION COEFFICIENTS

$$P(Y = 1 | x) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x)}}$$

Unlike SLR, the change in the predicted outcome is not the same for every fixed increase in x .



To make interpretation a little easier, we can isolate the **linear combination** $\theta_0 + \theta_1 x$.

ALGEBRA TIME: ISOLATE THE LINEAR COMBINATION $\theta_0 + \theta_1 x$

$$P(Y = 1 \mid x) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x)}}$$

$$1 + e^{-(\theta_0 + \theta_1 x)} = \frac{1}{P(Y = 1 \mid x)}$$

$$e^{-(\theta_0 + \theta_1 x)} = \frac{1}{P(Y = 1 \mid x)} - 1 = \frac{1 - P(Y = 1 \mid x)}{P(Y = 1 \mid x)}$$

$$e^{\theta_0 + \theta_1 x} = \frac{P(Y = 1 \mid x)}{1 - P(Y = 1 \mid x)}$$

$$\theta_0 + \theta_1 x = \log \left(\frac{P(Y = 1 \mid x)}{1 - P(Y = 1 \mid x)} \right)$$

INTERPRETING COEFFICIENTS JUST LIKE SLR, BUT WITH A NEW OUTCOME

$$P(Y = 1 | x) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x)}}$$

$$\log \left(\frac{P(Y = 1 | x)}{1 - P(Y = 1 | x)} \right) = \theta_0 + \theta_1 x$$

"For every one unit increase in x , $\log \left(\frac{P(Y = 1 | x)}{1 - P(Y = 1 | x)} \right)$ goes up by θ_1 ."

Is this an interpretable expression?

ODDS AND PROBABILITY

Does $\log \left(\frac{P(Y = 1 | x)}{1 - P(Y = 1 | x)} \right)$ have a meaningful interpretation?

Suppose someone says "Cal has **10 to 1 odds** of winning the game."

10:1 odds means that **$P(\text{Cal wins}) = 10 * P(\text{Cal loses})$** .

$$10 = \text{Odds}(\text{Cal wins}) = \frac{P(\text{Cal wins})}{P(\text{Cal loses})} = \frac{P(\text{Cal wins})}{1 - P(\text{Cal wins})}$$

$\frac{P(Y = 1 | x)}{1 - P(Y = 1 | x)}$ is the **odds** of the outcome $Y=1$ for a particular value of x .

INTERPRETING LOGISTIC REGRESSION COEFFICIENTS AS LOG ODDS

$$P(Y = 1 | x) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x)}}$$

$$\log \left(\frac{P(Y = 1 | x)}{1 - P(Y = 1 | x)} \right) = \theta_0 + \theta_1 x$$

"For every one unit increase in x , $\log \left(\frac{P(Y = 1 | x)}{1 - P(Y = 1 | x)} \right)$ goes up by θ_1 ."

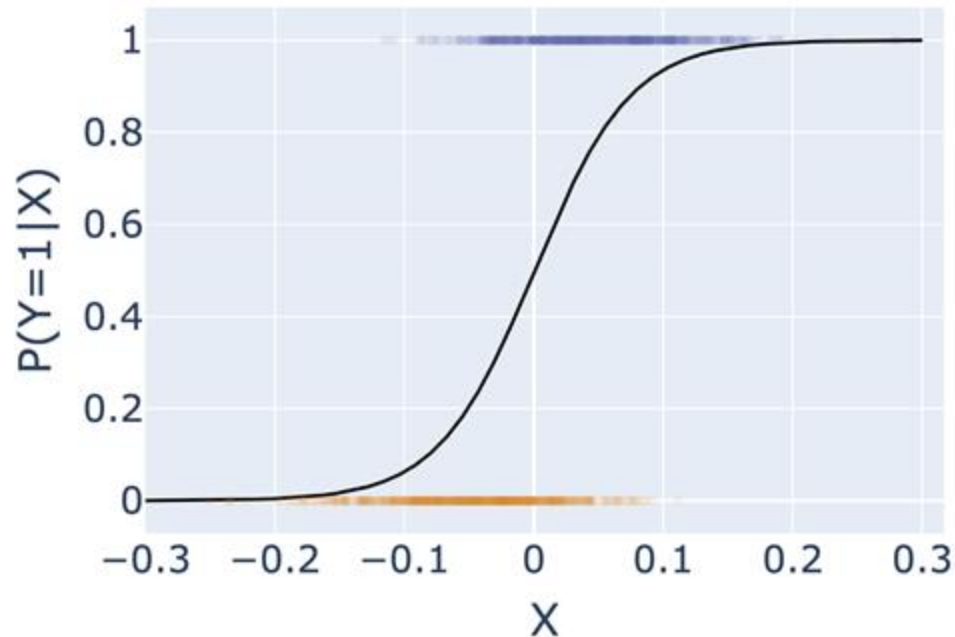
"For every one unit increase in x , **log(Odds(Y=1|x))** goes up by θ_1 ."

THERE IS LINEARITY IN THE LOGISTIC REGRESSION MODEL!

Probability is a **nonlinear** function of the features.

Probability is **not linear** in θ .

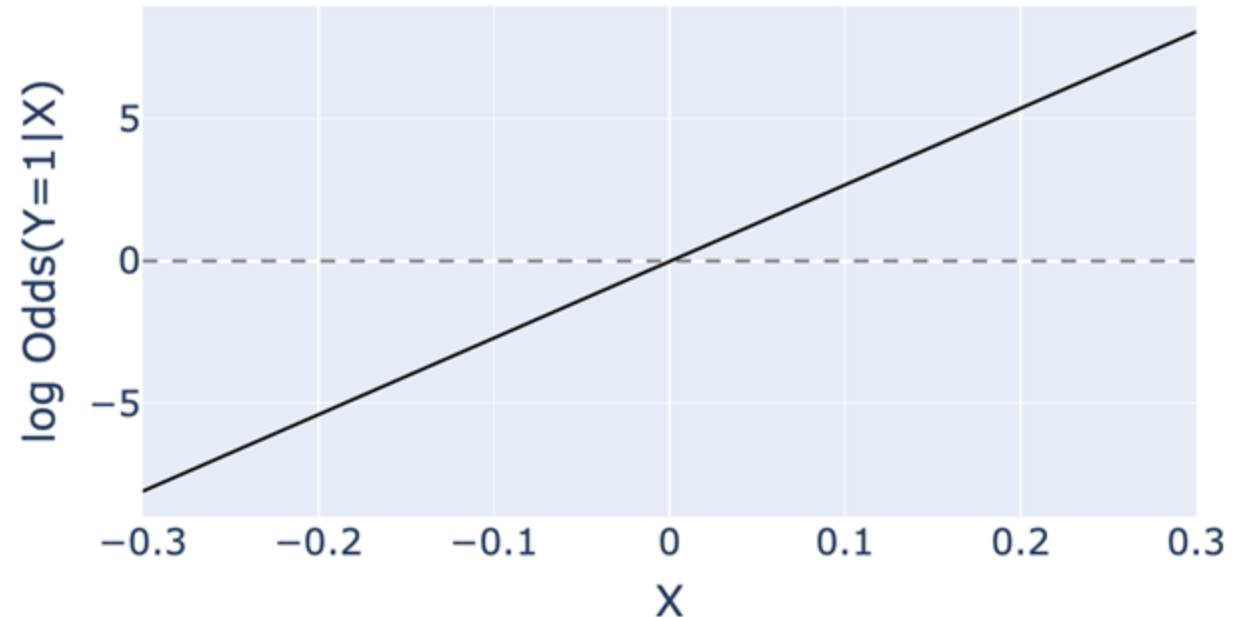
$$P(Y=1|x) = \sigma(\theta_0 + \theta_1 x)$$



Log odds are a **linear** function of the features.

Log odds are **linear** in θ .

$$\log(\text{Odds}(Y=1|x)) = \theta_0 + \theta_1 x$$



Two ways of showing exactly the same logistic regression model!

APPLYING A DECISION RULE

Logistic regression estimates the probability that a datapoint belongs to Class 1.

$$\hat{P}_{\theta}(Y = 1|x) = \frac{1}{1 + e^{-x^T\theta}}$$

The model itself does not **classify** (i.e, choose whether a prediction is Class 0 or Class 1).

A **decision rule** tells us how to convert model outputs into classifications. It's our decision.

THRESHOLDS

We commonly make decision rules by specifying a **threshold**, T . If the predicted probability is greater than or equal to T , predict Class 1. Otherwise, predict Class 0.

$$p = \hat{P}_\theta(Y = 1|x) = \frac{1}{1 + e^{-x^T \theta}}$$

$$\hat{y} = \text{classify}(x) = \begin{cases} \text{Class 1} & p \geq T \\ \text{Class 0} & p < T \end{cases}$$

The threshold T is often 0.5, but not always. We'll discuss using $T \neq 0.5$ next lecture.

CONVERTING PROBABILITIES TO PREDICTED CLASS

Suppose you have a fitted logistic regression model with the following form:

$$P(Y=1 | \mathbf{x}) = \sigma(1 + 2x_1 + 3x_2)$$

You decide to classify all new data points with $P(Y=1|\mathbf{x}) \geq 0.5$ as Class 1, and all points with $P(Y=1|\mathbf{x}) < 0.5$ as Class 0.

In other words, 0.5 is our chosen probability **threshold** for classification.

We emphasize this threshold is a choice. 0.5 is not necessarily the best option!

PROBABILITY OF 0.5 IMPLIES A LOG ODDS OF 0

Recall:

$$\log(\text{Odds}(Y=1 | \mathbf{x})) = \log \left(\frac{P(Y = 1 | x)}{1 - P(Y = 1 | x)} \right)$$

With $P(Y=1|x) = 0.5$, this becomes:

$$\log(\text{Odds}(Y=1 | \mathbf{x})) = \log \left(\frac{0.5}{1 - 0.5} \right) = \log \left(\frac{0.5}{0.5} \right) = \log(1) = 0$$

LINEARITY AND LOGISTIC REGRESSION

$$P(Y=1 | \mathbf{x}) = \sigma(1 + 2x_1 + 3x_2)$$

Equivalent way to write our model: $\log(\text{Odds}(Y=1 | \mathbf{x})) = 1 + 2x_1 + 3x_2$

You decide to classify all new data points with $P(Y=1|\mathbf{x}) \geq 0.5$ as Class 1, and all points with $P(Y=1|\mathbf{x}) < 0.5$ as Class 0.

This is the same as labeling all points with $\log(\text{Odds}(Y=1|\mathbf{x})) \geq 0$ as Class 1, and all points with $\log(\text{Odds}(Y=1|\mathbf{x})) < 0$ as Class 0.

$\log(\text{Odds}(Y=1|\mathbf{x})) = 0 = 1 + 2x_1 + 3x_2$ is a linear decision boundary for classification.

LOGISTIC REGRESSION PROVIDES A LINEAR DECISION BOUNDARY

θ_0 , θ_1 , and θ_2 from logistic regression model

$$0 = 1 + 2x_1 + 3x_2$$

Log odds threshold for classification.

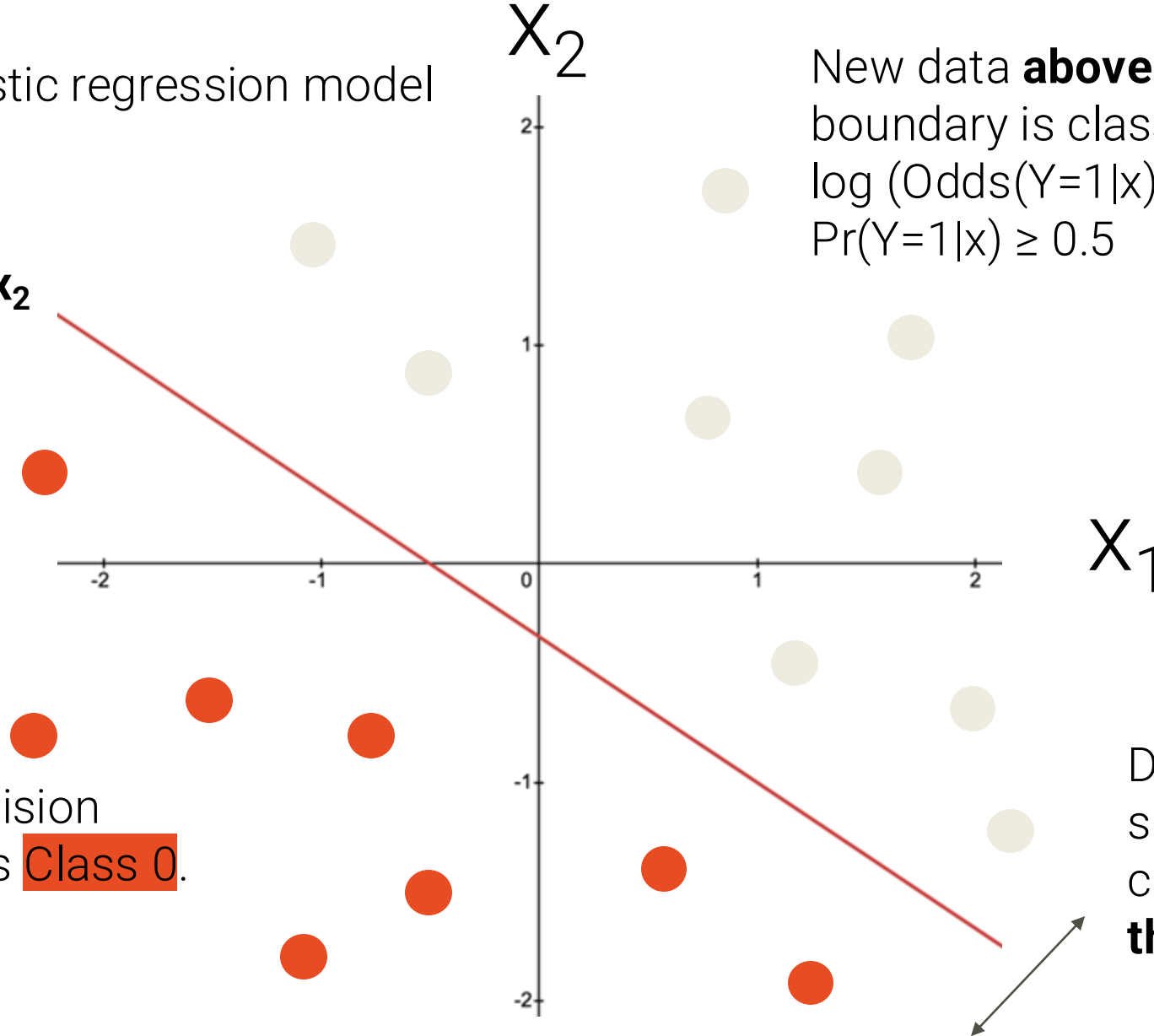
$$\log(\text{Odds}(Y=1|x)) = 0$$
$$\Pr(Y=1|x) = 0.5$$

New data **below** the decision boundary is classified as **Class 0**.

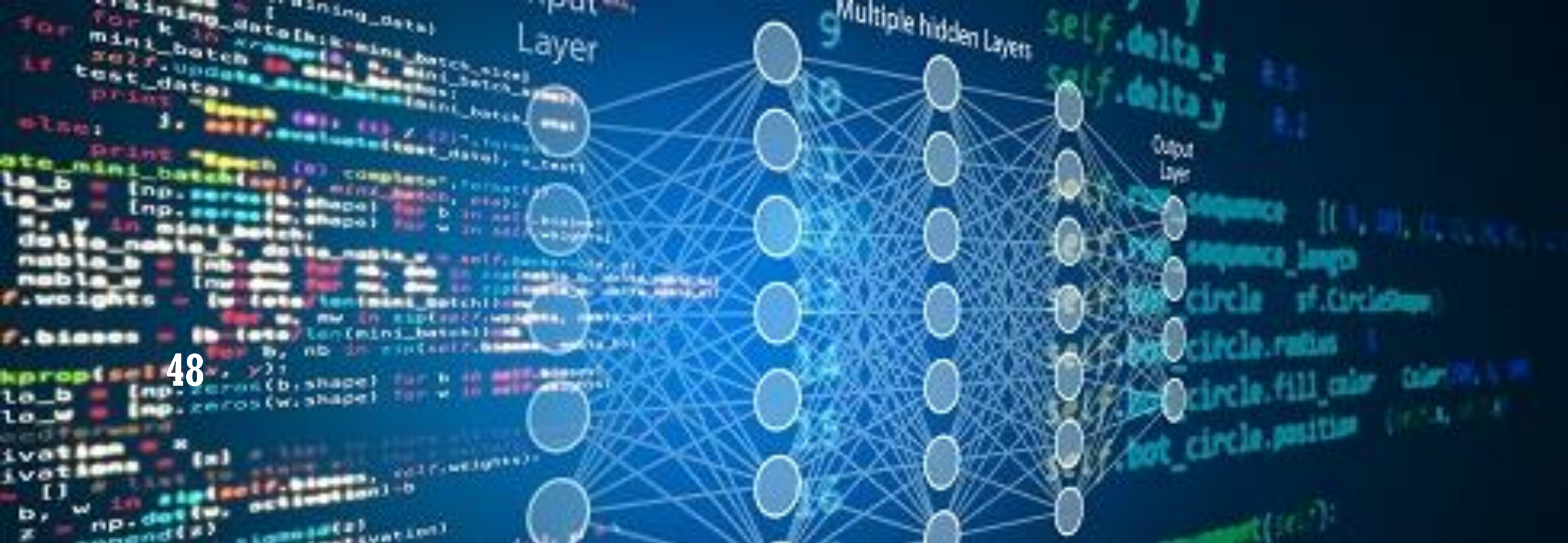
$$\log(\text{Odds}(Y=1|x)) < 0$$
$$\Pr(Y=1|x) < 0.5$$

New data **above** the decision boundary is classified as **Class 1**.

$$\log(\text{Odds}(Y=1|x)) \geq 0$$
$$\Pr(Y=1|x) \geq 0.5$$



Decision boundary shifts based on chosen classification **threshold**.



CROSS-ENTROPY LOSS

THE MODELING PROCESS

Regression ($y \in \mathbb{R}$)

1. Choose a model



Linear Regression

$$\hat{y} = f_{\theta}(x) = x^T \theta$$

2. Choose a loss function

Squared Loss or
Absolute Loss

3. Fit the model

Regularization
Sklearn/Gradient descent

4. Evaluate model performance

R^2 , Residuals, etc.

Classification ($y \in \{0, 1\}$)

Logistic Regression

$$\hat{P}_{\theta}(Y = 1|x) = \sigma(x^T \theta)$$

??

Regularization
Sklearn/Gradient descent

??

THE MODELING PROCESS

Regression ($y \in \mathbb{R}$)



1. Choose a model

Linear Regression

$$\hat{y} = f_{\theta}(x) = x^T \theta$$

2. Choose a loss function

Squared Loss or
Absolute Loss

3. Fit the model

Regularization
Sklearn/Gradient descent

4. Evaluate model
performance

R^2 , Residuals, etc.

Classification ($y \in \{0, 1\}$)

Logistic Regression

$$\hat{P}_{\theta}(Y = 1|x) = \sigma(x^T \theta)$$

Can squared (L2) loss still
work?

Regularization
Sklearn/Gradient descent

??

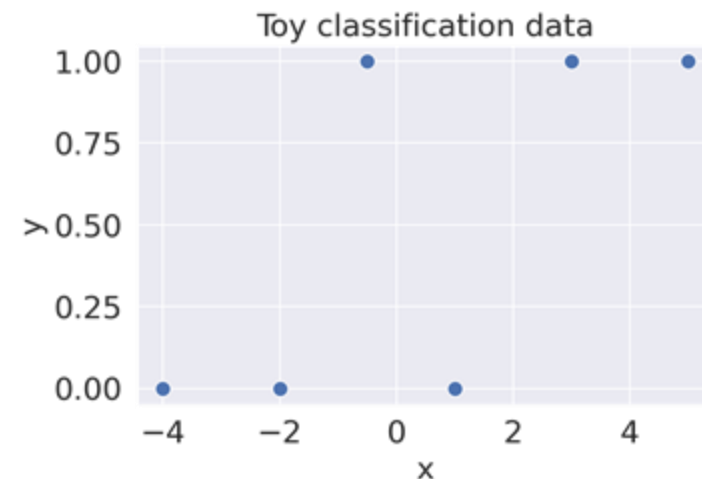
TOY DATASET: L2 LOSS AND COMPUTING OPTIMAL Θ

Logistic Regression model:

$$\hat{P}_{\theta}(Y = 1|x) = \sigma(x^T \theta)$$

Assume no intercept.
So x and θ are scalars.

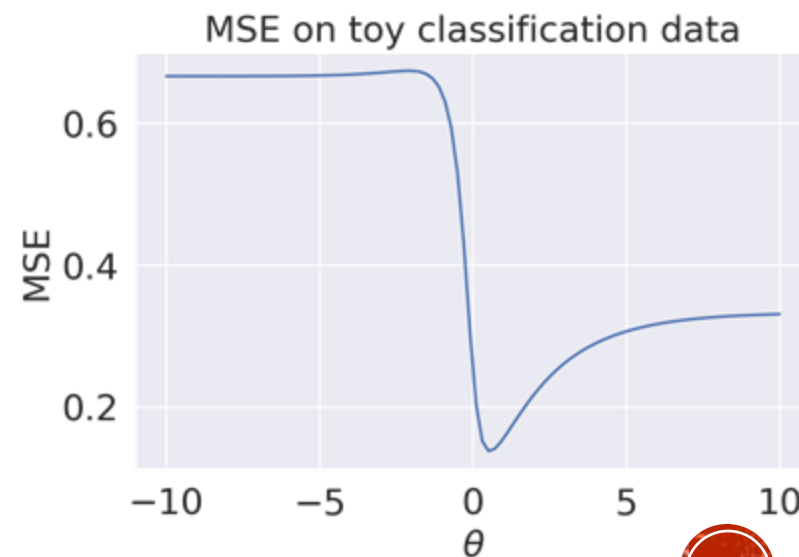
	x	y
0	-4.0	0
1	-2.0	0
2	-0.5	1
3	1.0	0
4	3.0	1
5	5.0	1



Mean Squared Error:

$$R(\theta) = \frac{1}{n} \sum_{i=1}^n (y_i - \sigma(x_i^T \theta))^2$$

The MSE loss surface for logistic regression has many issues!



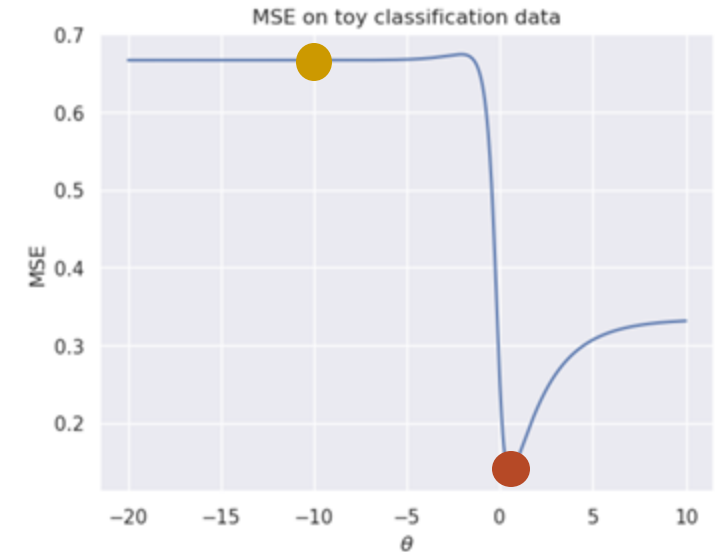
PITFALLS OF SQUARED LOSS

$$R(\theta) = \frac{1}{n} \sum_{i=1}^n (y_i - \sigma(x_i^T \theta))^2$$

1. Non-convex. Gets stuck in local minima.

Secant line crosses function, so $R''(\theta)$ is not greater than 0 for all θ .

Gradient Descent: Different initial guesses will yield different optimal estimates.



```
from scipy.optimize import minimize
```

```
minimize(mse_loss_toy_nobias, x0 = 0)["x"][0]
```

0.5446601825581691

```
minimize(mse_loss_toy_nobias, x0 = -5)["x"][0]
```

-10.343653061026611

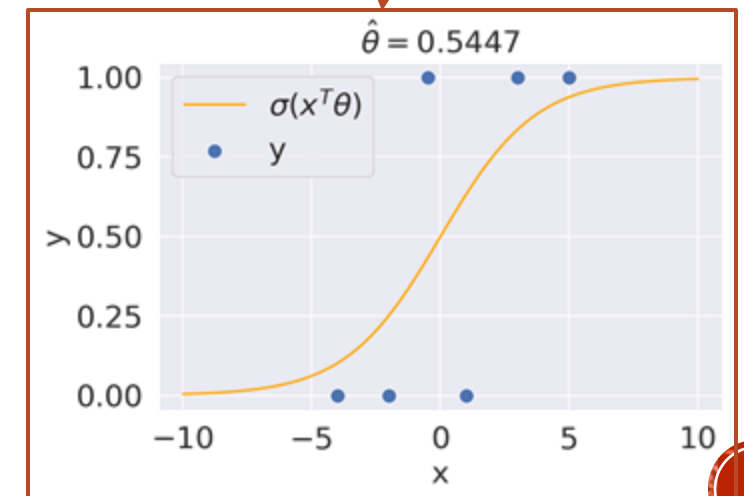
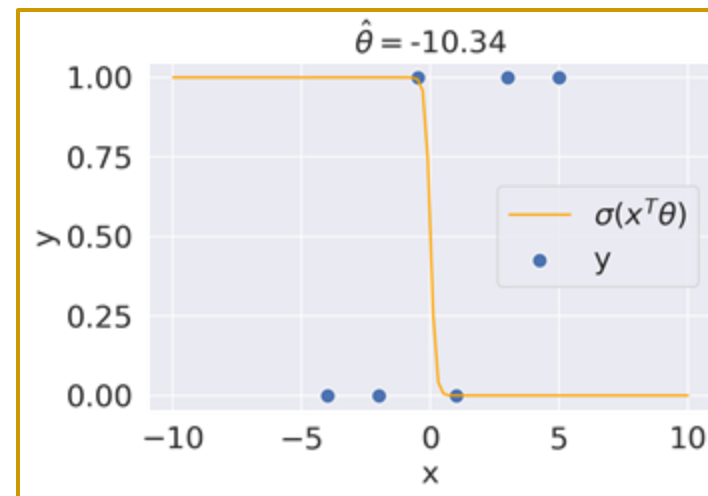
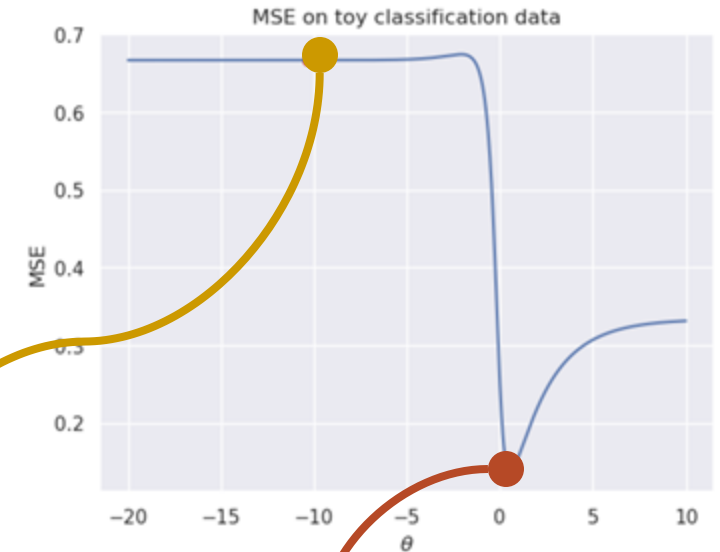
PITFALLS OF SQUARED LOSS

$$R(\theta) = \frac{1}{n} \sum_{i=1}^n (y_i - \sigma(x_i^T \theta))^2$$

1. Non-convex. Gets stuck in local minima.

Secant line crosses function, so $R''(\theta)$ is not greater than 0 for all θ .

Gradient Descent: Different initial guesses will yield different optimal estimates.



FOR BINARY CLASSIFICATION, L2 LOSS IS CAPPED AT 1

A logistic regression model outputs the estimated probability of belonging to Class 1.

What is the largest possible L2 loss for a single data point if we use logistic regression?

Solution:

Best case scenario:

- Estimate matches outcome. **L2 loss** = $(0-0)^2 = (1-1)^2 = 0$. Perfect prediction!

Worst case scenarios:

- Estimated probability = 0, Actual outcome = 1. **L2 loss** = $(0-1)^2 = 1$
- Estimated probability = 1, Actual outcome = 0. **L2 loss** = $(1-0)^2 = 1$

So, largest possible L2 loss is **1**. Not a very big penalty for a big error!

PITFALLS OF SQUARED LOSS

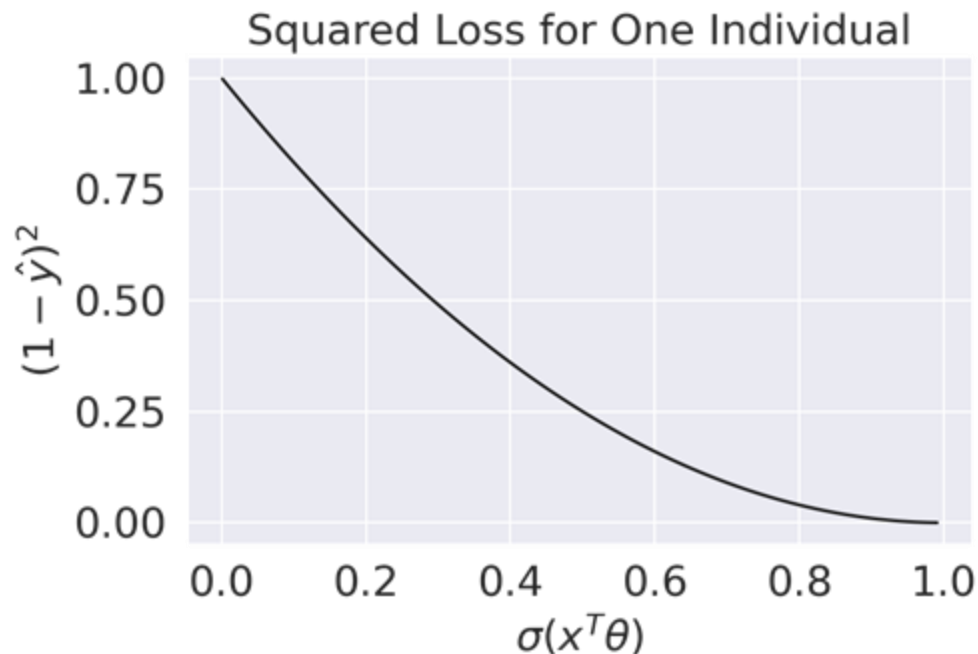
$$R(\theta) = \frac{1}{n} \sum_{i=1}^n (y_i - \sigma(x_i^T \theta))^2$$

1. **Non-convex.** Gets stuck in local minima.

2. **Bounded.** Not a good measure of model error.

⌘ We'd like loss functions to penalize “off” predictions.

⌘ MSE never gets very large, because both response and estimated probability are bounded by 1.



If true $y=1$ but
predicted
probability = 0:
 $(y - p)^2 = (1 - 0)^2 = 1$

CHOOSING A DIFFERENT LOSS FUNCTION

Regression ($y \in \mathbb{R}$)

Classification ($y \in \{0, 1\}$)



1. Choose a model

Linear Regression

$$\hat{y} = f_{\theta}(x) = x^T \theta$$

Logistic Regression

$$\hat{P}_{\theta}(Y = 1|x) = \sigma(x^T \theta)$$

2. Choose a loss function

Squared Loss or
Absolute Loss

Cross-Entropy Loss

3. Fit the model

Regularization
Sklearn/Gradient descent

Regularization
Sklearn/Gradient descent

4. Evaluate model
performance

R^2 , Residuals, etc.

??
(next time)

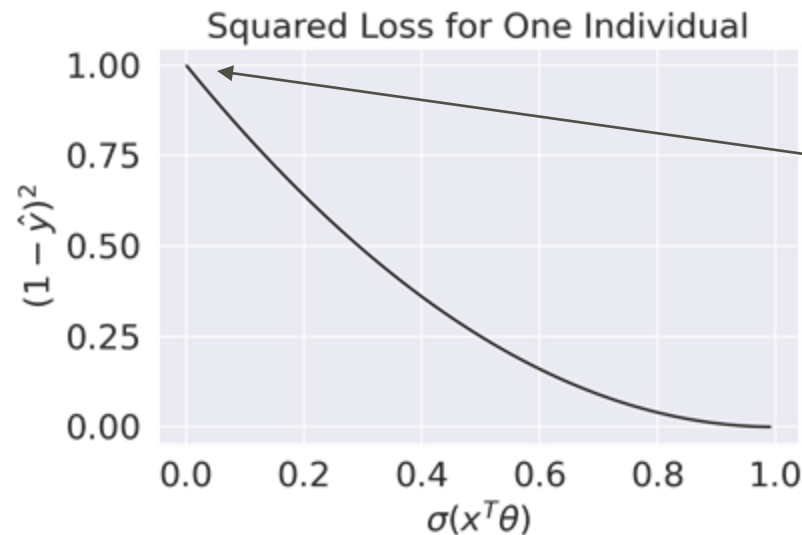
LOSS IN CLASSIFICATION

Let y be a binary label $\{0, 1\}$, and p be the model's estimated probability of Class 1.

In a classification task, how do we want our loss function to behave?

- When the true y is 1, we should incur **high loss** when the model estimates **low p** .
- When the true y is 0, we should incur **high loss** when the model estimates **high p** .

In other words, the behavior of our loss function depends on the value of the true class, y .



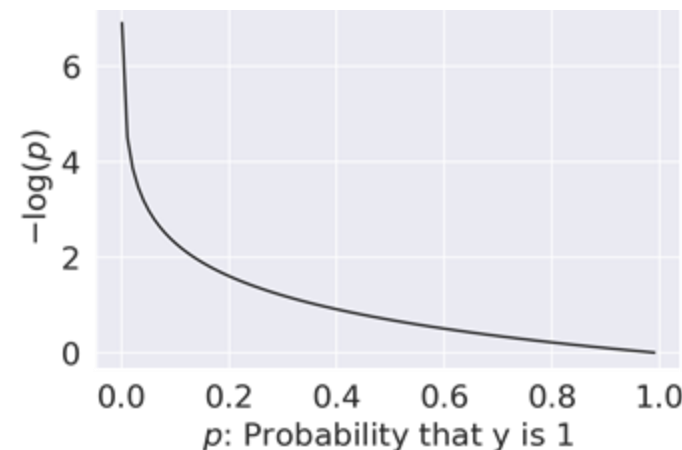
We want a loss function where this is a lot larger than 1!

CROSS-ENTROPY LOSS

Let y be a binary label $\{0, 1\}$, and p be the model's estimated probability of Class 1.

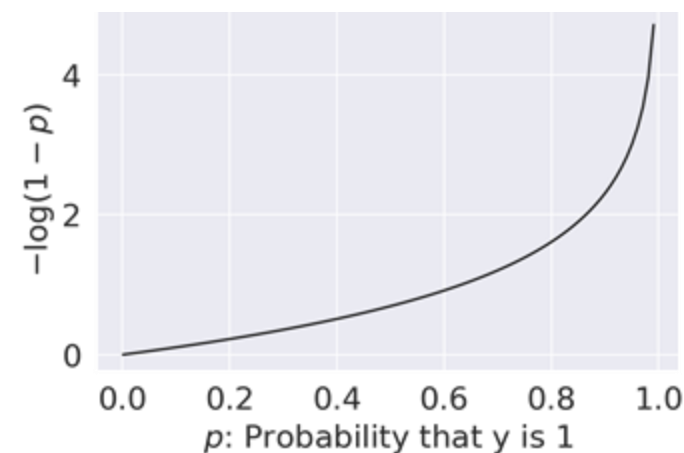
The **cross-entropy (CE) loss** is defined as:

$$\text{CE loss} = \begin{cases} -\log(p), & \text{if } y = 1 \\ -\log(1 - p), & \text{if } y = 0 \end{cases}$$



For $y = 1$,

- $p \rightarrow 0$: ∞ loss
- $p \rightarrow 1$: zero loss



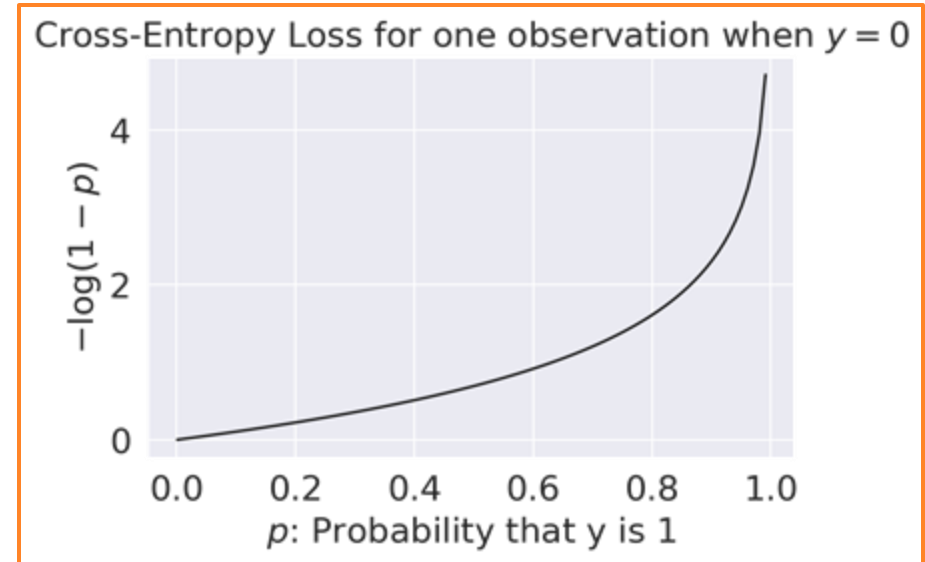
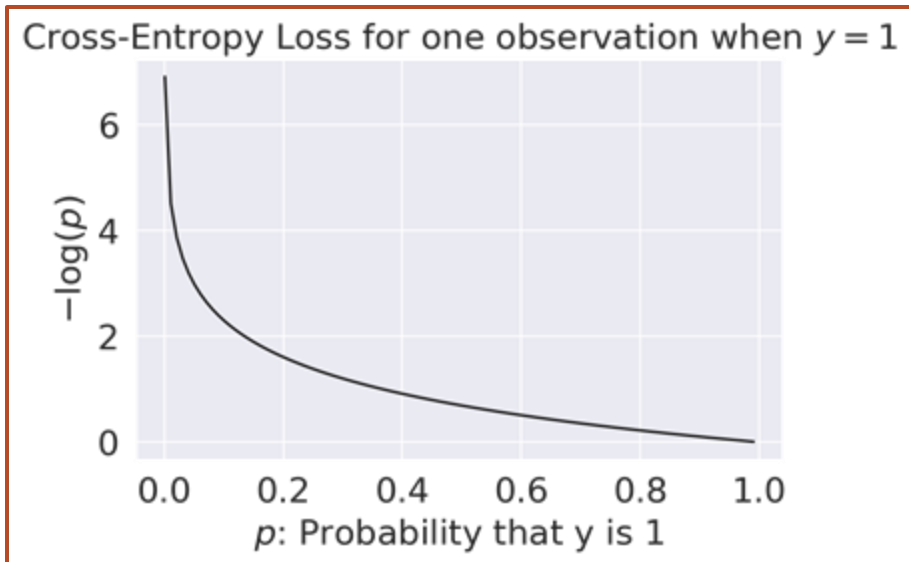
For $y = 0$,

- $p \rightarrow 0$: zero loss
- $p \rightarrow 1$: ∞ loss

CROSS-ENTROPY LOSS: TWO LOSS FUNCTIONS IN ONE!

The Cross-Entropy (CE) loss function is often written more compactly as the **product of the label and the log probability**, summed over both classes.

$$\text{CE loss} = \begin{cases} -\log(p), & \text{if } y = 1 \\ -\log(1 - p), & \text{if } y = 0 \end{cases} \rightarrow - \underbrace{(y \log(p))}_{\text{for } y = \mathbf{1}, \text{ only this term stays}} + \underbrace{(1 - y) \log(1 - p)}_{\text{for } y = \mathbf{0}, \text{ only this term stays}}$$



EMPIRICAL RISK: AVERAGE CROSS-ENTROPY LOSS

For a single datapoint, the cross-entropy (CE) loss is **convex**.

$$\text{CE loss} = - (y \log(p)) + (1 - y) \log(1 - p)$$

What about average CE loss, i.e., empirical risk?

$$p = \hat{P}_\theta(Y = 1|x) = \sigma(x^T \theta)$$

Empirical risk over a sample of size n , for logistic regression:

Estimated $P(Y=1|x)$ from logistic regression

$$\begin{aligned} R(\theta) &= -\frac{1}{n} \sum_{i=1}^n (y_i \log(p_i) + (1 - y_i) \log(1 - p_i)) \\ &= -\frac{1}{n} \sum_{i=1}^n (y_i \log(\sigma(\mathbb{X}_i^T \theta)) + (1 - y_i) \log(1 - \sigma(\mathbb{X}_i^T \theta))) \end{aligned}$$

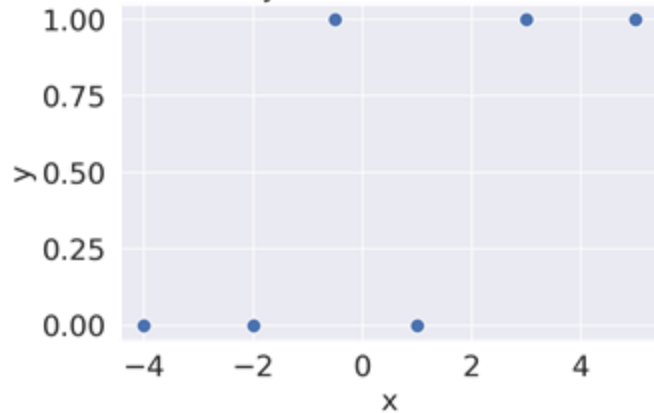
Optimization problem: Find the estimated $\hat{\theta}$ that minimizes $R(\theta)$:

$$\hat{\theta} = \arg \min_{\theta} -\frac{1}{n} \sum_{i=1}^n (y_i \log(\sigma(\mathbb{X}_i^T \theta)) + (1 - y_i) \log(1 - \sigma(\mathbb{X}_i^T \theta)))$$

CONVEXITY PROOF BY PICTURE

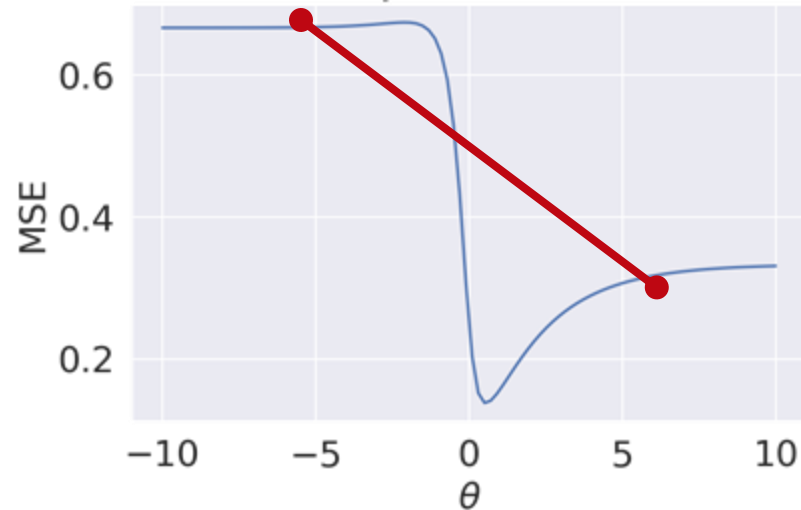
$$\hat{\theta} = \arg \min_{\theta} -\frac{1}{n} \sum_{i=1}^n (y_i \log (\sigma (\mathbb{X}_i^T \theta)) + (1 - y_i) \log (1 - \sigma (\mathbb{X}_i^T \theta)))$$

Toy classification data



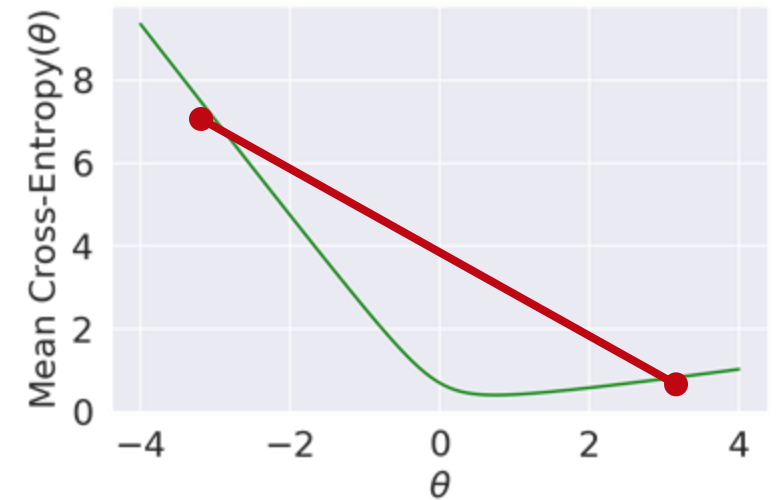
	x	y
0	-4.0	0
1	-2.0	0
2	-0.5	1
3	1.0	0
4	3.0	1
5	5.0	1

Squared Loss Surface



A straight line crosses the curve Non-convex

Cross-Entropy Loss Surface



Convex!

To be continued...

READING MATERIAL

- Chapter# 4: Principle of Data Science
- Chapter# 2 & 4: Hands-On ML [Aurélien]